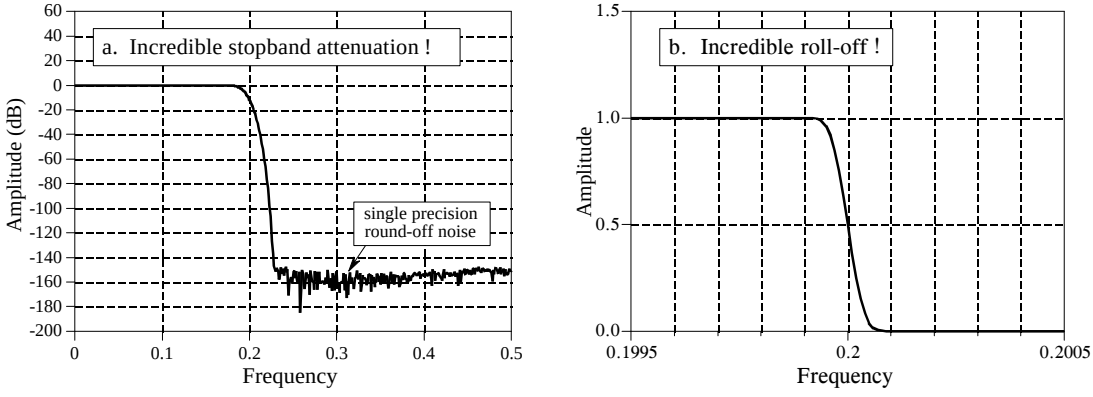

FIGURE 16-7
The incredible performance of the windowed-sinc filter. Figure (a) shows the frequency response of a windowed-sinc filter with increased stopband attenuation. This is achieved by convolving a windowed-sinc filter kernel with itself. Figure (b) shows the very rapid roll-off a 32,001 point windowed-sinc filter.

round-off noise on signals in the *passband* can erratically appear in the *stopband* with amplitudes in the -100dB to -120dB range).

Figure 16-7b shows another example of the windowed-sinc's incredible performance: a low-pass filter with 32,001 points in the kernel. The frequency response appears as expected, with a roll-off of 0.000125 of the sampling rate. How good is this filter? Try building an analog electronic filter that passes signals from DC to 1000 hertz with less than a 0.02% variation, and blocks all frequencies above 1001 hertz with less than 0.02% residue. Now that's a filter! If you really want to be impressed, remember that both the filters in Fig. 16-7 use *single precision*. Using *double precision* allows these performance levels to be extended by a *million* times.

The strongest limitation of the windowed-sinc filter is the *execution time*; it can be unacceptably long if there are many points in the filter kernel and standard convolution is used. A high-speed algorithm for this filter (FFT convolution) is presented in Chapter 18. Recursive filters (Chapter 19) also provide good frequency separation and are a reasonable alternative to the windowed-sinc filter.

Is the windowed-sinc the optimal filter kernel for separating frequencies? No, filter kernels resulting from more sophisticated techniques can be better. But beware! Before you jump into this very mathematical field, you should consider exactly what you hope to gain. The windowed-sinc will provide *any* level of performance that you could possibly need. What the advanced filter design methods may provide is a slightly shorter filter kernel for a given level of performance. This, in turn, may mean a slightly faster execution speed. Be warned that you may get little return for the effort expended.